Journal of Sports Sciences

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rjsp20

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To cite this article: Aaron M. Uthoff, Paul Macadam, James Zois, Ryu Nagahara, Jono Neville & John B. Cronin (2021): Effects of forearm wearable resistance during accelerated sprints: From a standing start position, Journal of Sports Sciences, DOI: 10.1080/02640414.2021.1943185

To link to this article: https://doi.org/10.1080/02640414.2021.1943185

Published online: 24 Jun 2021.

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JOURNAL OF SPORTS SCIENCES
https://doi.org/10.1080/02640414.2021.1943185

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Effects of forearm wearable resistance during accelerated sprints: From a standing start position

Aaron M. Uthoff[a], Paul Macadam[a], James Zois[b], Ryu Nagahara[c], Jono Neville[a] and John B. Cronin[a]

[a]Sports Performance Research Institute New Zealand (SPRINZ), AUT University, Auckland, New Zealand; [b]Faculty of Sports and Budo Coaching Studies, Institute for Health and Sport, Victoria University, Melbourne, Victoria, Australia; [c]National Institute of Fitness and Sports in Kanoya, Kanoya, Kagoshima, Japan

ABSTRACT

Fusiform weighted garments enable specific loading strategies during sport-specific movements. Loading the arms over during accelerated sprinting from a 2-point start position is pertinent to a variety of sporting performances. Fourteen sprint-trained individuals (age = 20.61 ± 1.16 years; height = 1.73 m ± 3.85 cm; body mass 65.33 ± 4.86 kg; personal best 100-m race time 11.40 ± 0.39 s) performed unloaded/loaded wearable resistance (WR) sprints. Between-condition step kinematics and kinetics were compared over four acceleration phases: steps 1–4, 5–8, 9–12 and 13–16. Sprint performance did not differ between unloaded and loaded WR at 10-m (−1.41%; ES = −0.32), or 30-m (−0.76%; ES = −0.24). Sprinting with forearm WR significantly decreased step frequency during phase two ($p < 0.05$, −3.42%; ES = −0.81) and three (−3.60%; ES = −0.86) and step velocity during phase four of the 30 m sprinting task ($p < 0.05$, −3.61%; ES: 0.91) only. There were no significant differences ($p \leq 0.05$) between step kinetics amongst the two conditions. Findings indicate that arm-loaded WR may provide specific sprinting overload for 2-point starting positions. This may be relevant to a wider sporting context such as field and team sport performances.

ARTICLE HISTORY
Accepted 10 June 2021

KEYWORDS
Wearable resistance; performance; athletes; training

Introduction

The utilization of forearm wearable resistance during block starts has emerged as a potential method to develop stride length and horizontal propulsion during the acceleration phase of 30 m block starts in sprint athletes (Uthoff et al., 2020). Investigating the application of these findings for use within a wider sporting context would require the utilization of a more specific sprinting performance which is initiated from a standing 2-point upright position. This would be of paramount importance to other sporting disciplines and provide practitioners with further capacity to implement such training tools into their training regimes.

Arm action has been proposed as a mechanism that functions to assist horizontal and vertical propulsion during accelerated and maximal velocity sprinting (Hinrichs et al., 1987; Schnier, 1982), both important determinants to team sport success. The number of accelerations performed during elite level male soccer has been linked to declines in soccer performance, when compared to high-speed running (Dalen et al., 2019). With 34% less accelerations demonstrated from the first minute compared to the last five-minutes of a soccer match (Dalen et al., 2019). Sprinting in team sports constitutes approximately 12% of total distance covered during soccer match play (Di Salvo et al., 2010), and in Australian Rules Football the number of accelerations completed by athletes is nearly doubled during the final periods (increased by 54%) when compared to the regular season of competition (Aughey, 2011). Previous research pertaining to Rugby athletes indicates the total number of accelerations, defined as movement >2.78 $m \cdot s^{-2}$, occur on average 78.30 occurrences during a rugby match (Glassbrook et al., 2019). Given these periods of acceleration occur at arguably the most important moments of team sport performance, and the recent findings suggesting beneficial effects of overloading the forearms during block starts (Uthoff et al., 2020), it seems relevant to investigate the effects of these training strategies during a 2-point standing position, which closer replicates the requirements of most physical activities completed in the above mentioned sports.

Knowing that force is the product of mass and acceleration, being able to perform specific movement patterns at high velocity through a full range of motion with added load, may provide an ideal method to optimize transference of training responses to explosive movements (Hrysomallis, 2012; Newton et al., 1999). One such method is the use of wearable resistance affixed to the arms. Developments in wearable resistance garments have enabled a greater variety of loading techniques with regard to positioning and magnitude for sprinting (for review, see Macadam, Cronin et al., 2018). However, compared to traditional resisted sprint training methods (Petrakos et al., 2016), acute adaptations to sprinting with wearable resistance affixed to the forearms is relatively unknown. Four studies currently provide the basis of our knowledge on the effects of loading the arms during over-ground sprinting. Sprinting performances up to 15 m while holding lead rods between 0.22 kg and 0.66 kg (0.60% to 1.80% body mass [BM]) reported no significant differences in acceleration ability (0%) as compared

CONTACT Aaron M. Uthoff James.Zois@vu.edu.au Sport Research Institute of New Zealand (SPRINZ), AUT University, 17 Antares Pl, Auckland 0632, New Zealand

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to unloaded sprinting in a group of recreationally trained males (Ropret et al., 1998). No changes were reported in 10-m and 40-m sprint performance when field-based athletes from sports disciplines such as field hockey and soccer performed over-ground sprinting with forearm wearable resistance of 0.50 kg on each arm (McNaughton & Kelly, 2010). In the third known study, 1 kg forearm loads which equated to ~2% BM was found to significantly decrease sprint performance over 10-m and 20-m (□2%) in a group of rugby players (Macadam, Simperingham et al., 2018). With the final study demonstrating 2% BM affixed to the forearms could help develop longer stride lengths, reduced stride frequency and increase vertical loading during 30 m sprints, from a block start position (Uthoff et al., 2020). With regard to kinematics, loads ≤0.66 kg have shown no effect on step characteristics (Ropret et al., 1998), while 1 kg loads have resulted in increased step length (2.10%) and contact time (6.54%) with accompanying decreases in step frequency (−4.13%) and flight time (−5.31%) over the first 5-m of accelerated sprinting (Macadam, Cronin et al., 2018).

Only two studies (Macadam, Cronin et al., 2018; Uthoff et al., 2020) have measured step kinetics while sprinting with forearm wearable resistance. Macadam, Simperingham et al. (Macadam, Cronin et al., 2018) identified that maximal horizontal power was 5.50% lower over the first 5-m than sprinting without load, while Uthoff et al. (Macadam, Cronin et al., 2018) identified a 4.12% increase in impulse during the late acceleration phase of a 30 m block start sprint, coinciding with a 7.70% increase in flight time which may have subsequently reduced stride frequency by −4.86%, in collegiate track sprinters wearing 2% BM forearm loads. These studies suggest that forearm wearable resistance loads ≥2% BM are required to significantly overload sprinting performance and associated step variables. The ability to train at high velocities with loads attached to distal segments of the body during sport-specific movement provides a highly specific loading strategy which may transfer to dynamic sporting actions, such as in-field sprinting performances. Considering the integral role of arm action for sprinting performance (Macadam, Cronin et al., 2018), the dearth of empirical reports on the acute effects of loading the arms during sprinting is surprising. Therefore, the purpose of this study was to determine the acute changes in step kinematic and kinetic characteristics when wearable resistance equivalent to 2% BM was attached to the forearms during short (30-m) over-ground accelerated sprinting tasks. Importantly, the current investigation is the first to research the effects of forearm loading during accelerated sprinting from a standing 2-point position, which provides broader application from a sporting context. Based on our understanding from the available literature, we hypothesize that sprinting from a 2-point start with 1% load on each forearm will result in longer stride lengths due to greater horizontal propulsive forces during early acceleration and decreased stride frequency as a result of longer flight times from increased vertical forces during mid-to-late acceleration phases.

Methods

To analyse the effects of forearm wearable resistance (FWR) on sprinting performance and step-related variables, a cross-sectional analysis of healthy collegiate male individuals was employed. Participants performed 30 m maximal effort accelerated sprints from a 2-point standing start with and without 2% body mass (BM) FWR across in-ground force plates. Step kinematic and kinetic variables were determined over four acceleration phases comprising the start (steps 1–4), early acceleration (steps 5–8), mid-acceleration (steps 9–12), and late acceleration (steps 13–16), based on distinct step characteristic for each phase (Nagahara et al., 2017a, 2017b; Zois et al., 2014). The study assessed how adding 2% BM FWR could influence underlying and overt mechanics across multiple phases of sprint acceleration.

Participants

Fourteen recreational track and field male athletes (age: 20.61 ± 1.16 years; body mass: 65.33 ± 4.86 kg; height: 1.73 m ± 3.85 cm; personal best 100 m race time: 11.40 ± 0.39 s) volunteered to participate in this study. All participants were informed about the objective and potential risks and benefits of the study prior to giving their written consent. None of the participants reported any musculo-skeletal injuries prior to, or during the study. All athletes were familiar with the WR garments and had worn these before in prior training sessions. This study was conducted in accordance with the Declaration of Helsinki, with ethics approval provided by the University ethics board.

Procedures

The testing was conducted on an indoor track with Mondo track surface. Each participant completed his own individual warm-up under his coach's supervision comprising progressive running drills interspersed with dynamic stretching and sub-maximal runs ranging from 50% to 100% of maximal effort. Thereafter, participants performed three maximal effort sprints from a 2-point standing start over 30 m: two were performed with a 2% BM forearm load and one sprint which was conducted un-loaded; 0% additional BM (UL). We performed two loaded attempts to ensure maximum attempts were achieved. Both FWR sprints were completed in sequence either before or after the UL sprint, depending on each participant's individual randomized order of testing. No potentiation effect (Zois et al., 2014) was observed with FWR when completed prior to the UL trial. Trials were separated by 5 min of passive recovery. Athletes performed all trials wearing track spike shoes and tight-fitting clothing. The one UL trial, and the fastest FWR trials were used for subsequent data processing and analysis.

Equipment

The WR garments worn by the participants during the loaded condition were Lila™ Exogen™ compression sleeves (Sportboleh Sdh Bhd, Malaysia). This WR garment allowed for loads of 0.55 kg to 0.75 kg with Velcro backing, to be attached to compressive sleeves on each arm. The WR load of 2% BM was distributed between both arms, and loads were evenly placed in a dove tailed manner around the circumference of the forearm (Figure 1). Loads of 2% BM were used in this study as prior

Figure 1. Forearm wearable resistance load distribution.

WR arm studies have reported significant performance changes with this magnitude (Macadam, Simperingham et al., 2018).

Step kinematics and kinetics were quantified using 30 in-ground force plates (TF-90,100, Tec Gihan, Uji, Japan; 1000 Hz) which quantified ground reaction force (GRF) data (Nagahara et al., 2017a, 2018). An electronic starting gun connected to an operating computer provided the synchronization for the timer and force plates. The electronic starting system also started a photo-cell timing system (TC Timing System, Brower, Draper, UT, USA) which quantified 0–10, 10–30 and 0–30 m sprint times.

Data analyses

Step-to-step kinematic variables over 30 m were generated from GRF data. Contact time (CT) and flight time (FT) were considered when vertical GRF rose above and fell below 20 N in accordance with previous studies (Nagahara et al., 2017a, 2017b, 2018). Using the inverse of step duration, step frequency (SF) was calculated from the foot strike of one leg, to the foot strike of the other leg. The distance between two contralateral foot strikes determined step length (SL). The centre of pressure in the anterior-posterior direction during the middle of the contact phase determined foot position at each step (Nagahara et al., 2017a). Time integration of three-dimensional GRFs was used to calculate vertical impulse (IMP_V), net mediolateral impulse (IMP_{ML}), propulsive impulse (IMP_{H+}), braking impulse (IMP_{H-}) and net horizontal impulse (IMP_{H+} minus IMP_{H-}; IMP_H) over the 30 m distance. Mean horizontal propulsive force ($F\chi$) was computed by averaging the propulsive force at each step. The kinetic variables were normalized to the corresponding individual body mass to determine how much force is relatively produced per kg of body mass. In order to eliminate the influence of random noise and accurately detect the ground contact and flight phases of the step cycle, GRF signals were filtered using a Butterworth low-pass digital filter at a cut-off frequency of 50 Hz (Nagahara et al., 2017a, 2018).

To gain insights into whether the variables were influenced by WR forearm loading at different phases of accelerated sprinting, we determined average values for every four steps from the 1st to the 16th similar to Girard, Brocherie, Morin, Degache, and Millet (Girard et al., 2015) and Nagahara, et al. (Nagahara et al., 2018). Due to 16 steps being the smallest number taken over the 30 m sprint for all participants, we standardized the number of steps to be analysed to 16. Accordingly, representing each variable during accelerated sprinting up to 16 steps was quantified over four acceleration phases (1st to 4th steps, 5th-8th steps, 9th-12th steps and 13th-16th steps). Note that steps 1–4, steps 5–8, steps 9–12 and steps 13–16 will be referred to as acceleration phases 1, 2, 3 and 4, respectively, from this point forward.

Statistical analysis

Analyses were completed using Excel (Microsoft Excel 2016, version 16.13.1, Microsoft Corporation, Redmond, WA) and SPSS 25.0 statistical software (SPSS, Chicago, IL). Typical descriptive statistics (means and standard deviations) were calculated for all statistical comparisons. The Shapiro–Wilk statistic was used to check the data for normal distribution. A one-way analysis of variance (ANOVA) was used to compare sprint performances between the loading conditions over 0–10-m and 0–30-m. A 2 × 4 mixed model ANOVA using repeated measures was used to analyse the between-condition step kinematic and kinetic variables across loaded and unloaded conditions over the four acceleration phases (i.e., 1, 2, 3 and 4) with Bonferroni post hoc comparisons. Effect sizes (ES) between loaded and unloaded conditions were evaluated using Cohen's *d* (Cohen, 1988). Percentage changes were calculated using the average individual change, and practical importance of ES was evaluated as <0.2 (trivial); 0.2–0.6 (small); 0.6–1.2 (moderate); 1.2–2.0 (large); >2.0 (very large) (Hopkins, 2017). An alpha level of $p \leq 0.05$ was set and 95% confidence intervals (CI) were used.

Results

Sprinting performance, as measured by time to complete the 30 m all out task was not significantly altered when comparing the UL and FWR groups (see Table 1). However, the addition of FWR did cause some significant changes in kinematic variables.

Kinematic and kinetic variables

As was step frequency was significantly reduced across phases 2 and 3 when FWR was implemented ($p < 0.05$, −3.41%; ES: 0.81

Table 1. Changes in 30 m sprint times associated with and without forearm wearable resistance.

Variable	UL M ± SD	FWR M ± SD	% Change (95% CI)	ES (95% CI)
10 m sprint (s)	2.22 ± 0.10	2.26 ± 0.09	−1.41 (−2.68 to −0.15)	−0.32 (−2.19 to 1.54)
30 m sprint (s)	4.60 ± 0.15	4.64 ± 0.14	−0.76 (−1.43 to −0.10)	−0.24 (−1.53 to 1.05)

N.B. UL unloaded; FWR forearm wearable resistance; M mean; S seconds; SD standard deviation; CI confidence interval; *d* Cohen's d effect size.

and −3.60%; ES: 0.86, respectively). Small to moderate non-significant increases were reported for contact time (ES = 0.28–0.65) across all the 4 sprinting phases, as was flight time during phases 2 and 4 (ES = 0.46–0.60, respectively), when athletes wore FWR. Small increases in step length were observed over all phases of sprinting performance within the FWR group (ES = 0.40–0.52). Table 2 further depicts the changes in kinematic variables over the 30 m sprinting task. There was no significant difference when comparing kinetic variables across the two groups; however, small to moderate differences were reported for vertical impulse across all four phases of the task ranging from 5.09% (ES = 0.54) in phase 1 to 5.64% (ES: 0.69) in phase 4. This also coincided with small reductions in mean propulsive force during the early acceleration phase of −3.52% (ES = 0.33) within the FWR group. Further comparisons of practical importance can be seen in Table 3.

Discussion

The current investigation is the first to elucidate the application of FWR during an accelerated sprint from a starting position, broadening our understanding of WR application to other sporting disciplines, such as team sport for example. The major findings of this investigation indicate that wearing FWR of 2% BM does not severely impede overall sprinting performance, as derived by time to completion. The addition of FWR did, however, provide some alterations to kinematic and kinetic variables measured during the 30 m sprinting trials, which would be of interest to practitioners aiming to implement a sport-specific overload strategy.

Sprint performance was not significantly hindered over either 10 m or 30 m sprint distances when using the FWR. However, there were small effects which may suggest slower sprint times within the FWR group compared to unloaded. These were reported at both distances of 10 m (−1.41%: ES = 0.32) and at 30 m (−0.76%; ES = 0.24), although neither of these were statistically significant ($P > 0.05$). Our findings are not dissimilar to research investigating the use of 2% FWR across similar distances but from a block starting position, more akin to track athletes ($P > 0.05$; 1.38% to 1.75%: ES = −0.38 to −0.54) (Uthoff et al., 2020). Similarly, others using lesser loads (0.66 kg forearm loading) during sprinting performances in trained hockey, soccer and distance running athletes (McNaughton & Kelly, 2010; Ropret et al., 1998) have also reported no change in sprint time variables. Intuitively, practitioners may consider that increasing loads would negatively affect overall time to completion of sprint and acceleration tasks. However, it seems plausible that other kinematic variables may be further recruited, or relied upon, to compensate for the additional load and hence maintaining all-out sprint performance across the 30 m task. This may include greater recruitment of prime motor units, or perhaps improved synchronization during this complex springing task which could aid in the overall performance. This would be of great interest to conditioning practitioners attempting to identify a method of sport-specific overload which can target fundamental step kinematics (Hunter et al., 2005; Nagahara et al., 2018) during sprinting tasks, whilst not affecting overall technique.

Indeed, results from the current investigation support the above hypothesis to some extent, with significant decreases reported in the FWR group for the kinematic variable of step frequency (2nd and 3rd phases). Small to moderate changes between groups may further shed light on the utilization of FWR during the standing sprinting tasks, with FWR increasing contact time (ES = 0.28–0.65, across all phases), flight time (ES = 0.46–0.60 only phases 2–4) and step length (ES = 0.40–0.52, across all phases) over the 30 m, 2-point standing sprint. The link between step frequency and sprint performance has been previously purported with a range of track athletes including non-specialist, national and world class sprinters (Morin et al., 2015). Similarly, others have noted that higher running speeds are achieved via greater step velocity and a reduction in overall ground reaction times (McMahon & Green, 1979; Weyand et al., 2010, 2000). Therefore, overloading step velocity and step frequency using FWR could provide athletes with a tool to further enhance these qualities via regular training programmes. Whilst further research is required to elucidate this hypothesis, it has potentially important implications for individuals looking to implement a more specific means to overload the acceleration phase of sprinting, create appropriate overload without compromising skill execution of the task of interest, and maximize training time with the potential use of FWR during other skilled tasks such as match play or small sided game activity.

The mechanisms responsible for spatiotemporal changes during accelerated FWR sprinting may be further described via kinetic analyses given that kinematics changes result from underlying kinetic processes (Moir, 2015). For example, the ability to generate high vertical and resultant ground reaction forces has been linked with greater step lengths (Lockie et al., 2013). In the current investigation, the introduction of FWR reported small to trivial decreases in mean propulsive force (ES = −0.33), increases in propulsive impulse (ES = 0.22–0.23) and increases in net anterior-posterior impulse during some of the acceleration phases (ES = 0.20–0.24). While small to moderate increases in vertical impulse was observed across all the four phases (ES = 0.54–0.69) when wearing the FWR tool (see Table 3). These findings are similar to those of Uthoff et al. (Uthoff et al., 2020), who discovered that ~2% BM FWR resulted in small to moderate changes in mean propulsive force (ES = 0.41–0.66) and propulsive impulse (ES = 0.30–1.03) during the first two acceleration phases, small increases in net anterior-posterior impulse towards the end of the acceleration (ES = 0.26), and small to moderate increases in vertical impulse across all acceleration phases (ES = 0.31–0.89). Combined, these results suggest that the implementation of FWR during accelerated sprints from either a 2-pont standing

Table 2. Step kinematics over 30 m sprinting with and without forearm wearable resistance.

	Phase 1			Phase 2			Phase 3			Phase 4		
Variable	UL M ± SD	FWR M ± SD	ES (95% CI)	UL M ± SD	FWR M ± SD	ES (95% CI)	UL M ± SD	FWR M ± SD	ES (95% CI)	UL M ± SD	FWR M ± SD	ES (95% CI)
Step velocity (m/s)	5.64 ± 0.26	5.64 ± 0.24	−0.03 (−0.77 to 0.71)	7.53 ± 0.26	7.51 ± 0.27	−0.06 (−0.80 to 0.68)	8.44 ± 0.27	8.41 ± 0.31	−0.10 (−0.84 to 0.64)	8.85 ± 0.29	8.80 ± 0.32	−0.16 (−0.90 to 0.59)
Step frequency (Hz)	4.21 ± 0.24	4.08 ± 0.20	−0.60 (−1.99 to 0.79)	4.39 ± 0.18	4.24 ± 0.17*	−0.81 (−2.21 to 0.59)	4.44 ± 0.19	4.28 ± 0.18*	−0.86 (−2.25 to 0.53)	4.35 ± 0.24	4.19 ± 0.19	−0.72 (−2.11 to 0.66)
Step length (m)	1.34 ± 0.11	1.39 ± 0.10	0.40 (−2.00 to 2.80)	1.72 ± 0.12	1.77 ± 0.12	0.47 (−1.80 to 2.73)	1.91 ± 0.12	1.97 ± 0.12	0.52 (−1.51 to 2.54)	2.04 ± 0.15	2.10 ± 0.12	0.45 (−1.49 to 2.41)
Step width (m)	0.30 ± 0.06	0.29 ± 0.06	−0.12 (−5.22 to 4.97)	0.22 ± 0.07	0.20 ± 0.05	−0.30 (−6.00 to 5.39)	0.14 ± 0.07	0.13 ± 0.05	−0.22 (−7.74 to 7.31)	0.11 ± 0.06	0.11 ± 0.04	−0.04 (−8.34 to 8.25)
Contact time (s)	156.2 ± 13.0	164.8 ± 13.6	0.65 (−.42 to 0.87)	128.8 ± 5.86	134.0 ± 7.33	0.43 (0.20 to 0.66)	112.8 ± 6.99	114.7 ± 7.03	0.28 (0.01 to 0.54)	107.7 ± 6.26	109.6 ± 5.81	0.31 (0.04 to 0.58)
Flight time (s)	82.7 ± 10.9	81.6 ± 12.4	−0.10 (−0.40 to 0.21)	103.7 ± 10.7	108.4 ± 9.88	0.46 (0.17 to 0.75)	113.2 ± 11.0	119.7 ± 10.8	0.60 (0.33 to 0.86)	123.1 ± 13.2	129.6 ± 10.10	0.55 (0.30 to 0.80)

N.B. UL un-loaded sprinting; FWR forearm wearable resisted sprinting; M mean; SD standard deviation; CI confidence interval; * $p < 0.05$.

Table 3. Step kinetics over 30 m sprinting with and without forearm wearable resistance.

	Phase 1			Phase 2			Phase 3			Phase 4		
Variable	UL M ± SD	FWR M ± SD	ES (95% CI)	UL M ± SD	FWR M ± SD	ES (95% CI)	UL M ± SD	FWR M ± SD	ES (95% CI)	UL M ± SD	FWR M ± SD	ES (95% CI)
Mean propulsive force (N)	368.3 ± 39.2	355.3 ± 39.1	0.33 (−1.08 to 0.42)	305.7 ± 35.9	303.6 ± 36.7	0.06 (−0.80 to 0.68)	306.5 ± 40.4	313.7 ± 37.3	0.19 (−0.56 to 0.93)	307.2 ± 37.6	306.4 ± 33.7	−0.02 (−0.76 to 0.72)
Braking impulse (N·s)	−2.25 ± 0.92	−2.13 ± 0.93	0.12 (−0.62 to 0.86)	−4.82 ± 1.29	−4.60 ± 1.20	0.18 (−0.56 to 0.92)	−7.91 ± 1.54	−8.32 ± 1.97	−0.23 (−0.97 to 0.51)	−11.2 ± 2.21	−11.8 ± 2.03	−0.32 (−1.06 to 0.43)
Propulsive impulse (N·s)	52.3 ± 5.42	53.5 ± 5.23	0.22 (−0.52 to 0.96)	30.0 ± 2.99	29.9 ± 3.09	−0.04 (−0.78 to 0.70)	22.3 ± 2.34	22.8 ± 2.30	0.23 (−0.51 to 0.97)	19.2 ± 1.98	19.2 ± 2.01	0.02 (−0.72 to 0.76)
Net AP impulse (N·s)	54.6 ± 5.95	55.6 ± 5.25	0.19 (−0.56 to 0.93)	34.9 ± 3.95	34.5 ± 4.12	−0.08 (−0.82 to 0.66)	30.2 ± 3.70	31.1 ± 4.03	0.24 (−0.50 to 0.99)	30.3 ± 3.62	32.2 ± 2.24	0.20 (−0.54 to 0.94)
Net ML impulse (N·s)	19.9 ± 5.64	21.5 ± 6.00	0.27 (−0.48 to 1.01)	12.6 ± 5.25	12.6 ± 3.77	0.01 (−0.73 to 0.75)	8.49 ± 0.98	8.81 ± 2.47	0.03 (−0.72 to 0.77)	7.71 ± 1.54	8.01 ± 1.71	0.01 (−0.73 to 0.75)
Effective max vertical force (N)	1509.1 ± 140.4	1496.3 ± 177.3	−0.08 (−0.82 to 0.66)	1879.0 ± 193.0	1907.0 ± 204.1	0.14 (−0.60 to 0.88)	2101.2 ± 243.9	2195.0 ± 267.4	0.37 (−0.38 to 1.11)	2302.9 ± 296.1	2400.6 ± 301.2	0.33 (−0.42 to 1.07)
Vertical impulse (N·s)	149.2 ± 14.0	156.8 ± 14.5	0.54 (−0.22 to 1.29)	146.0 ± 11.8	153.4 ± 13.0	0.60 (−0.16 to 1.35)	145.6 ± 11.7	153.9 ± 12.8	0.67 (−0.09 to 1.44)	148.7 ± 12.4	157.1 ± 11.8	0.69 (−0.07 to 1.45)

N.B. UL Un-resisted sprinting; FWR forearm wearable resisted sprinting; M mean; SD standard deviation; CI confidence interval; *d* Cohen's d effect size.

or block start could be an appropriate tool for overloading specific kinetic mechanisms responsible for optimal sprinting performance in non-track event sprinting tasks. This may be achieved via augmenting force orientation during the early phases of accelerated sprinting or overloading the vertical application of force, particularly over the latter phases of the sprint acceleration.

Limitations

Future research in WR should incorporate training studies with elite and sub-elite populations to identify the long-term adaptations of WR within both participant categories. This will aid to establish the potential use of WR as an ergogenic aid in athletic development. The use of force plates with greater sampling rates may provide researchers with more detailed information; however, this was not achievable as the current plates were built into the unique 30-m track. Finally, investigating the kinematic interaction between the arms and legs during loaded WR would be of high importance to elucidate the entire overload contribution, when the limbs are loaded during sprinting performance. Perhaps there is an optimal individual WR load required to achieve an effective overload of the arms, this is yet to be established.

Conclusions and future research

The current investigation contributes to the notion of training specificity using FWR in acceleration tasks which occur from a standing position. These findings suggest that 2% BM overload can be applied to the upper limbs with limited detriments to sprinting technique and subsequent performance, which should be reassuring to practitioners. Identifying specific training strategies which improve step-related characteristics and arm action during sprinting may assist practitioners in programming for improved sprint performances and training adaptations. The cumulative effect of the changes in force application resulting in the expression of longer contact times, flight times and step length, with a reduction in step frequency, provides practitioners with a sound basis for implementing FWR as part of their training strategy to overload, and subsequently enhance accelerated sprinting performances.

Disclosure of potential conflicts of interest

Professor John Cronin holds the position of Head of Research at Lila Movement Technology, the manufacturer of the wearable resistance garment used in this study. However, Professor Cronin is an academic researcher first and was blinded from the data collection and analysis.

ORCID

Aaron M. Uthoff http://orcid.org/0000-0002-6737-0562
Paul Macadam http://orcid.org/0000-0002-2077-5386
James Zois http://orcid.org/0000-0001-5025-3119
Ryu Nagahara http://orcid.org/0000-0001-9101-9759
John B. Cronin http://orcid.org/0000-0002-8889-0911

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